DSA0405 - Fundamentals of Data Science

Total points 21/25

CO2-AT3 Assessment Tool MCQs

✓ 1. A dataset follows a **normal distribution** with **μ = 500** and **σ = 50**. What is the probability that a randomly chosen value is **above 600**? 1/1

- (●) a) 2.28% ✓
- ( ) b) 5%
- ( ) c) 16%
- ( ) d) 0.13%

✓ 2.A dataset contains salaries that follow a **normal distribution** with **μ = 75,000** and **σ = 10,000**. 1/1

What is the **Z-score** for a salary of **95,000**?

- ( ) a) 1.5
- (●) b) 2.0 ✓
- ( ) c) 2.5
- ( ) d) 3.0

3.A Hadoop cluster stores **500 GB** of data with a **block size of 128 MB** and a **replication factor of 3**. 1/1

How much **total storage** is required in the cluster?

- a) 500 GB
- b) 1000 GB
- c) 1500 GB
- d) 2000 GB

4.A dataset contains **1 million records**, and **20% of them** belong to **Class A**. If we randomly **sample 5000 records**, what is the expected number of Class A records in the sample? 1/1

- a) 1000
- b) 1250
- c) 1500
- d) 2000

5. A MapReduce job processes **1 TB** of data using **100 nodes**. If each node processes **10 GB per hour**, how long will the entire job take? 1/1

- a) 1 hour
- b) 2 hours
- c) 10 hours
- d) 20 hours

6.A dataset has **1000 samples**, where:
600 belong to Class A 400 belong to Class B

0/1

A split results in two child nodes:
Left: **300 Class A, 50 Class B** Right: **300 Class A, 350 Class B**

What is the **information gain**?

- a) 0.125
- b) 0.278
- c) 0.541
- d) 0.672

Correct answer

- c) 0.541

7. A dataset has **10 features**. PCA retains **95% variance** with the top **4 components**. What percentage of variance do the remaining components contribute?

1/1

- a) 2%
- b) 5%
- c) 10%
- d) 20%

8. A dataset contains values ranging from **1 to 1,000,000**. Which transformed value does **10,000** map to if we apply a **log transformation using base 10**? 1/1

- a) 2
- b) 3
- c) 5
- d) 4

9.A dataset follows a **normal distribution** with **μ = 150** and **σ = 20**. What percentage of values lie **beyond 3 standard deviations** from the mean (both tails combined)? 1/1

- a) 0.3%
- b) 1.35%
- c) 2.5%
- d) 5%

10.A dataset has **1 million records** and **5% of them** contain at least **one missing value**. If we remove **all rows** with missing values, how many records remain? 1/1

- a) 900,000
- b) 950,000
- c) 990,000
- d) 975,000

11. A classification model gives **Precision = 0.80** and **Recall = 0.60**. What is the **F1-score**? 1/1

- a) 0.65
- b) 0.70 ✓
- c) 0.75
- d) 0.80

12. Consider the following NumPy matrix: 1/1

A=[2 4 3 6]

- a) 0 ✓
- b) 2
- c) 6
- d) 12

13.A **Pandas DataFrame** with **1 million rows** is sorted using df.sort_values(by='column1'). The sorting algorithm used by Pandas is: 1/1

- a) Merge Sort
- b) Quick Sort
- c) Tim Sort ✓
- d) Heap Sort

14. A Matplotlib plot renders **1 million data points** using plt.scatter(). The plot is slow. What is the best way to optimize performance? 1/1

- a) Use plt.plot() instead of plt.scatter()
- b) Enable marker="." in plt.scatter()
- c) Use ax.hexbin() instead of plt.scatter() ✓
- d) Increase figure size

15. A **Pandas DataFrame** has **10 million rows** and **100 columns**. Which method is **fastest** for selecting **specific rows** based on conditions? 0/1

- a) df[df['column1'] > 100]
- b) df.loc[df['column1'] > 100] ✗
- c) df.query('column1 > 100')
- d) df.apply(lambda x: x['column1'] > 100, axis=1)

Correct answer

- c) df.query('column1 > 100')

16. A **NumPy array** of shape **(100,000, 10)** is stored using **float64** data type. What is the total memory consumption in MB? 1/1

- a) 8 MB
- b) 80 MB
- c) 160 MB ✓
- d) 320 MB

17. A **K-Means clustering model** is run on **1 million points** with **K=5**. The convergence time is **10 minutes**. If **K is doubled to 10**, assuming all other factors remain the same, what is the expected increase in execution time? 1/1

- a) No change
- b) 2x increase ✓ (selected)
- c) 4x increase
- d) 10x increase

18.A **NumPy** array of shape **(200,000, 5)** is stored using **float32** instead of **float64**. How much memory is saved in MB? 0/1

- a) 2 MB
- b) 4 MB
- c) 8 MB
- d) 16 MB ✗ (selected)

Correct answer

- c) 8 MB

19. A dataset contains **100,000 data points**, and a **scatter plot** is generated using **Matplotlib**. Each marker takes **100 bytes** of memory. What is the total memory usage for the scatter plot? 1/1

- a) 1 MB
- b) 5 MB
- c) 10 MB ✓ (selected)
- d) 100 MB

20. A **Pandas DataFrame** with **1 million rows** is sorted based on a **single column**. The sorting algorithm used by Pandas is **Timsort (O(n log n))**. If sorting takes **5 seconds**, how long would it take for **10 million rows**? 1/1

- a) 10 seconds
- b) 50 seconds
- c) 25 seconds
- d) 60 seconds

21. A Pandas DataFrame contains **20,000 rows**. After applying a filter that removes **40% of the data**, how many rows remain? 1/1

- A) 10,000
- B) 12,000
- C) 14,000
- D) 16,000

22. A NumPy array contains **numbers from 1 to 500**. What is the mean of this dataset? 1/1

- • A) 245
- • B) 250
- • C) 255
- • D) 275

23. A dataset contains **5000 values** distributed into **50 bins** in a histogram. What is the average number of values per bin? 1/1

- A) 80
- B) 90
- C) 100 (selected) ✓
- D) 110

24.Which sub-package of SciPy is specifically designed for handling vector quantization and K-Means clustering algorithms? 0/1

- A) scipy.optimize
- B) scipy.spatial (selected) ✗
- C) scipy.cluster
- D) scipy.linalg

Correct answer

- C) scipy.cluster

25.What core data science task are both NumPy and SciPy explicitly built to solve together? 1/1

- A) Web scraping and API deployment.
- B) Mathematical and numerical analysis.
- C) Static data visualization and dashboarding.
- D) Database management and query optimization. (selected) ✓